

Department of Computer Sciences

CS-1005- Discrete Structures

Semester: Fall 2024

Assignment #1

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Due Date: As mentioned on Google Classroom

Max Score 150

Note:

1. Please submit handwritten solutions to all of the questions given below.
2. Scan your handwritten solution and upload to Google class room.
3. No extension in deadline.
4. Plagiarism would result in zero marks.

Question 1: Propositional Calculus (1 hour)

1. Basic Logical Connectives (6 marks)

- A. Express the following statements using logical connectives:
 1. If it rains, the ground gets wet.
 2. The lights are off or the door is closed.
 3. It is not true that both the switch is on and the light is off.
- B. Write each of these propositions in the form “p if and only if q” in English.
 1. If it is hot outside you buy an ice cream cone, and if you buy an ice cream cone it is hot outside.
 2. For you to win the contest it is necessary and sufficient that you have the only winning ticket.
 3. The trains run late on exactly those days when I take it.

2. Truth Tables (12 Marks)

- A. Construct truth tables for the following compound propositions:
 1. $(P \wedge Q) \rightarrow R$
 2. $\neg(P \vee Q) \leftrightarrow (\neg P \wedge \neg Q)$
 3. $(p \vee q) \rightarrow (p \oplus q)$

3. Logical Equivalence (10 marks)

- A. Show that $(p \rightarrow q) \vee (p \rightarrow r)$ and $p \rightarrow (q \vee r)$ are logically equivalent without using truth table.
- B. Show that $p \leftrightarrow q$ and $\neg p \leftrightarrow \neg q$ are logically equivalent without using truth table.

Question 2: Predicates and Quantifiers (1.5 hours)

1. Expressing Statements with Quantifiers (6 marks)

- A. Translate the following statements into logical expressions using quantifiers:
 1. All students in the class are intelligent.
 2. Some people love discrete mathematics.
 3. There exists a number that is both even and prime.

2. Negation of Quantified Statements (6 marks)

A. Negate the following statements:

1. $\forall x(P(x) \rightarrow Q(x))$
2. $\exists y(P(y) \wedge \neg Q(y))$

3. Converting Negations of the statements (6 marks)

A. Write the formal negation for each of the following statements. Show the process of computing the negation.

1. All dogs are friendly.
2. Some estimates are accurate.
3. Every valid argument has a true conclusion.

4. Combining Quantifiers (4 marks)

A. Determine the truth value of the following statements (Assume the domain of discourse is the set of all natural numbers):

1. $\forall x \exists y \ x+y=10$
2. $\exists x \forall y \ x \times y=0$

5. Nested Quantifiers (6 marks)

A. Translate the following statements into formal logical expressions:

1. Every even number is greater than some odd number.
2. There exists a person who knows everyone.

Question 3: Inference Rules (2 hours)

1. What rule of inference is used in each of these arguments? (15 marks)

1. Ali is a mathematics major. Therefore, Ali is either a mathematics major or a computer science major.
2. Jehanzeb is a mathematics major and a computer science major. Therefore, Jehanzeb is a mathematics major.
3. If it is rainy, then the pool will be closed. It is rainy. Therefore, the pool is closed.
4. If it snows today, the university will close. The university is not closed today. Therefore, it did not snow today.
5. If I go swimming, then I will stay in the sun too long. If I stay in the sun too long, then I will sunburn. Therefore, if I go swimming, then I will sunburn.

B. For each of these sets of premises, what relevant conclusion or conclusions can be drawn? Explain the rules of inference used to obtain each conclusion from the premises.

1. "I am either dreaming or hallucinating." "I am not dreaming." "If I am hallucinating, I see elephants running down the road."
2. "I am either clever or lucky." "I am not lucky." "If I am lucky, then I will win the lottery."
3. "All insects have six legs." "Dragonflies are insects." "Spiders do not have six legs." "Spiders eat dragon- flies."

C. For each of these arguments determine whether the argument is correct or incorrect and explain why.

1. All students in this class understand logic. Saleem is a student in this class. Therefore, Saleem understands logic.

2. Every computer science major takes discrete mathematics. Natasha is taking discrete mathematics. Therefore, Natasha is a computer science major.
 3. All parrots like fruit. My pet bird is not a parrot. Therefore, my pet bird does not like fruit.
2. **Prove that if n is an integer and $3n+2$ is even, then n is even using (10 marks)**
- A. using proof by contraposition.
 - B. using proof by contradiction.

Question 4: Set Operations (1 hour)

1. Suppose a list A contains the 30 students in a mathematics class, and a list B contains the 35 students in an English class, and suppose there are 20 names on both lists. Find the number of students: **(12 marks)**
 1. only on list A
 2. only on list B
 3. on list A or B (or both),
 4. on exactly one list.
2. Suppose a list A contains 40 students in a Mathematics class, a list B contains 50 students in an English class, and a list C contains 45 students in a Physics class. If there are 20 students on both lists A and B , 15 students on both lists B and C , and 10 students on both lists A and C , with 5 students on all three lists: **(15 marks)**
 1. Find the number of students only on list A .
 2. Find the number of students only on list B .
 3. Find the number of students only on list C .
 4. Find the total number of students on at least one of the three lists.
 5. Find the number of students on exactly two of the lists.
3. In a school, there are three clubs: Debate Club, Chess Club, and Drama Club. The Debate Club has 25 members, the Chess Club has 30 members, and the Drama Club has 35 members. If 10 students are in both Debate and Chess, 12 are in both Chess and Drama, and 8 are in both Debate and Drama, with 5 students in all three clubs: **(12 marks)**
 1. How many students are in only one club?
 2. How many students are in at least one of the three clubs?
 3. How many students are in exactly two clubs?
 4. If 10 students are not in any club, how many students are there in total?

Question 5: Functions (30 minutes)

1. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = 3x - 2$. **(6 marks)**
 - a) Prove that $f(x)$ is an onto function.
 - b) Modify the function to make it no longer onto and explain why the modified function is not onto.
2. Given the following functions: $f(x) = 2x^2 + 3x - 5$] [$g(x) = \sqrt{x+7}$] **(12 marks)**
 - a) Find the composite functions $((f \circ g)(x))$ and $((g \circ f)(x))$.
 - b) Determine the domain and range of $((f \circ g)(x))$.
 - c) Let $(h(x) = f^{-1}(x))$. Find $(h(x))$ and verify that $(h(f(x)) = x)$.
3. Let $f: \mathbb{Z} \rightarrow \mathbb{Z}$ be defined by $f(x) = x^3 - x$. **(12 marks)**
 - a) Prove or disprove that $f(x)$ is an onto function.
 - b) Determine if $f(x)$ is an injective (one-to-one) function.
 - c) If $f(x)$ is bijective, find its inverse function.